
Customer Churn Prediction & Retention Analytics

An end-to-end machine learning project on the Telco Customer Churn dataset: exploratory analysis, predictive modeling, business-oriented threshold optimization, and retention strategy recommendations.

Dataset	IBM Telco Customer Churn (7,043 customers, 21 features)
Models evaluated	Logistic Regression, Random Forest, XGBoost
Selected model	Logistic Regression (ROC-AUC 0.842)
Business threshold	0.30 predicted-probability cutoff (recall-oriented)
Report scope	Data cleaning → EDA → modeling → explainability → retention strategy

Data source: IBM / Kaggle Telco Customer Churn dataset (WA_Fn-UseC_-Telco-Customer-Churn.csv). All figures and metrics in this report are taken directly from the accompanying analysis notebook.

Executive Summary

This project analyzes customer churn for a telecommunications provider using a dataset of 7,043 customers. Approximately **26.5% of customers churned**, establishing a meaningful class imbalance that shaped both the modeling and evaluation approach. Exploratory analysis identified four consistent drivers of churn: short customer tenure, month-to-month contracts, fiber-optic internet service, and electronic-check payment.

Three classification models were trained and compared — Logistic Regression, Random Forest, and XGBoost. **Logistic Regression achieved the strongest overall performance** (ROC-AUC of 0.842) and was selected as the primary model, with its coefficients used to explain churn drivers in business terms. Because the retention use case prioritizes catching potential churners, the default 0.50 classification threshold was replaced with a **0.30 threshold**, raising recall from 55.9% to 75.4%. Customers were then segmented into Low, Medium, and High risk tiers to support prioritized retention outreach.

This report also documents the modeling methodology and its current limitations — most notably that the decision threshold was tuned on the test set rather than a separate validation set — so that results are interpreted with appropriate caution and next steps are clear.

Key Numbers at a Glance

Metric	Value
Overall churn rate	26.54% (1,869 of 7,043 customers)
Best model (ROC-AUC)	Logistic Regression — 0.842
Recall at business threshold (0.30)	75.4%
High-risk customers identified (test set)	209 of 1,409 (14.8%)
Highest-churn segment	Month-to-month contracts — 42.71%

1. Project Overview & Objective

Customer churn — the loss of subscribers to a competitor or to cancellation — directly affects recurring revenue, and acquiring a new customer typically costs substantially more than retaining an existing one. The objective of this project is to build a predictive model that identifies customers at high risk of churning *before* they leave, and to translate the model's output into a practical risk-segmentation and retention framework.

The analysis follows five stages:

1. **Data cleaning** — resolving invalid values and confirming data quality.
2. **Exploratory data analysis** — identifying which customer attributes are associated with churn.
3. **Predictive modeling** — training and comparing Logistic Regression, Random Forest, and XGBoost.
4. **Threshold optimization** — tuning the classification cutoff to match the business objective.
5. **Explainability & strategy** — translating model coefficients into risk segments and retention actions.

2. Dataset & Data Preparation

The dataset contains **7,043 customer records** and 21 attributes spanning demographics (gender, senior citizen status, partner/dependents), account information (tenure, contract type, payment method, billing charges), and subscribed services (phone, internet, streaming, security add-ons). The target variable, *Churn*, indicates whether the customer left within the observed period.

Data cleaning steps applied:

TotalCharges contained blank values for 11 customers with zero tenure (new customers with no billing history yet); these were converted to numeric and set to 0.

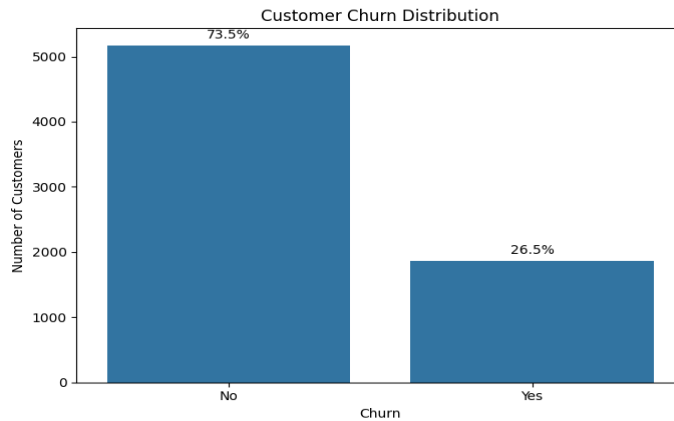
No missing values remained after this correction, and no duplicate customer IDs were found.

Categorical fields were checked for inconsistent or unexpected category labels; none were found.

3. Exploratory Data Analysis

The goal of this stage was to understand which customer characteristics are associated with churn, in order to guide feature use and provide business context ahead of modeling.

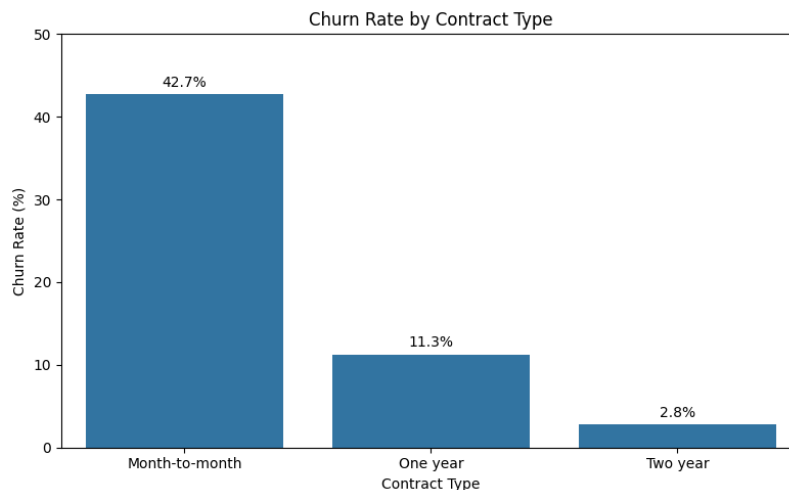
3.1 Overall Churn Distribution



BUSINESS INSIGHT — Class Imbalance

Approximately 26.5% of customers in the dataset churned. This imbalance means model performance should not be evaluated on accuracy alone — a model that always predicts “No Churn” would already reach roughly 73.5% accuracy without identifying a single at-risk customer.

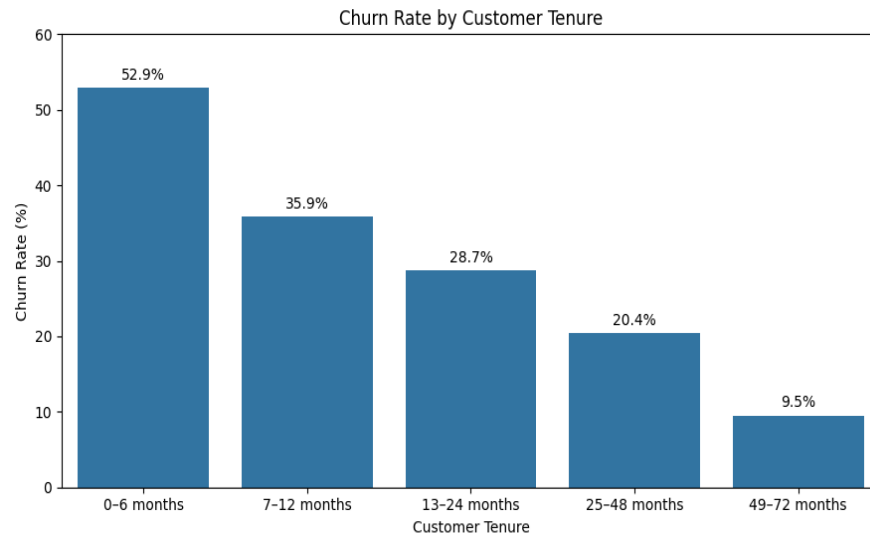
3.2 Churn by Contract Type



BUSINESS INSIGHT — Contract Length Matters

Month-to-month customers churn at 42.71%, versus 11.27% for one-year and just 2.83% for two-year contracts — a strong association between contract flexibility and attrition risk, though this descriptive analysis alone does not establish that contract length causes lower churn.

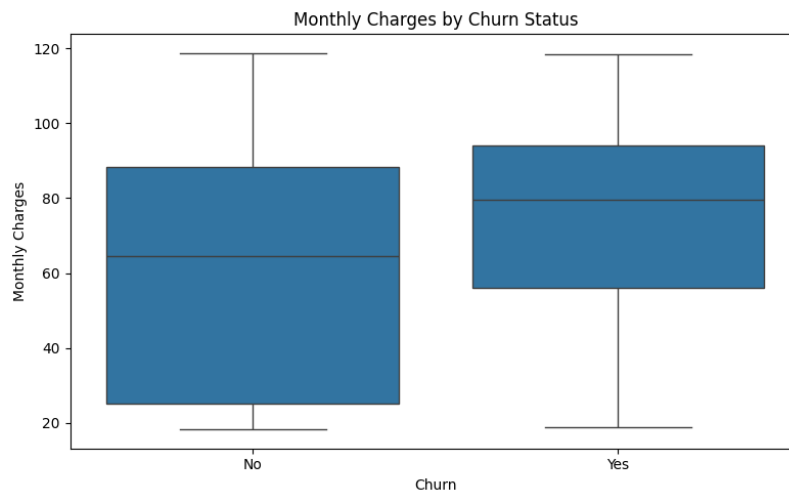
3.3 Churn by Customer Tenure



BUSINESS INSIGHT — Early Lifecycle Risk

Churn is highest among newer customers: 52.94% in the first 6 months versus 9.51% for customers with 49–72 months of tenure. The early relationship period appears to be the highest-risk window, suggesting onboarding and early engagement as priority levers.

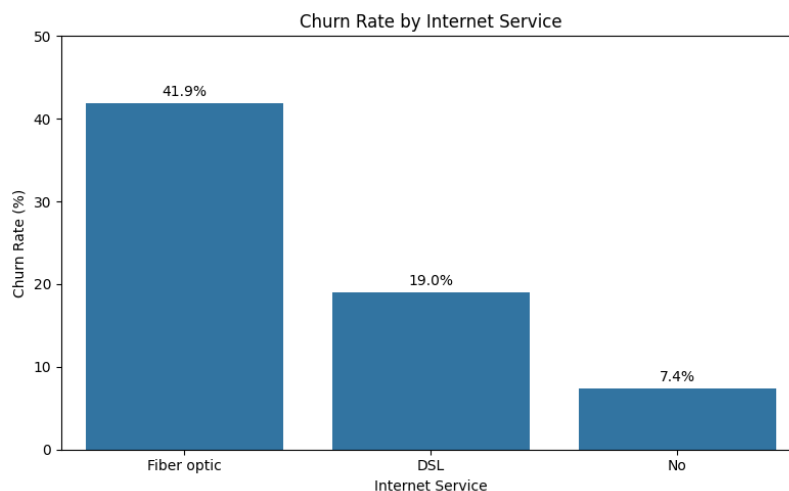
3.4 Monthly Charges by Churn Status



BUSINESS INSIGHT — Pricing Signal

Churned customers had a higher median monthly charge (\$79.65) than retained customers (\$64.43). The distributions overlap substantially, though, so monthly charges alone do not cleanly separate churners from non-churners.

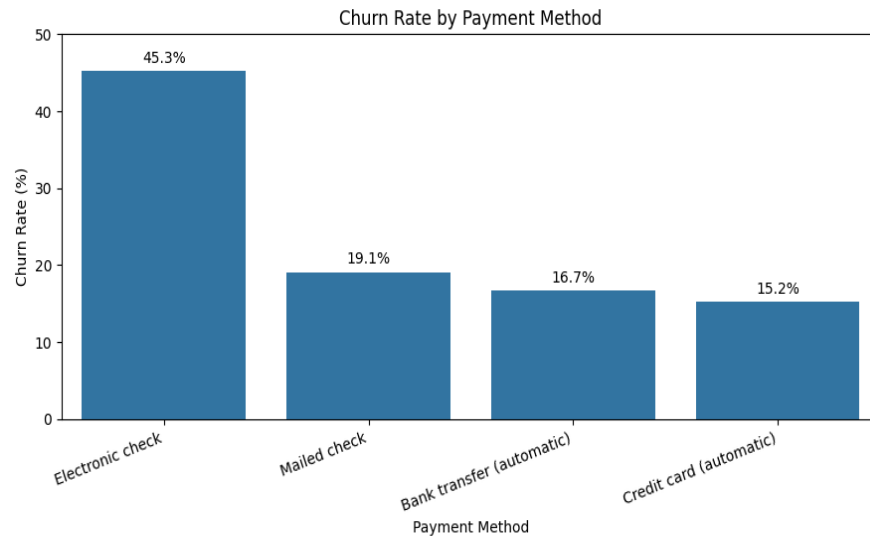
3.5 Churn by Internet Service



BUSINESS INSIGHT — Fiber Optic Segment

Fiber-optic customers churn at 41.89%, well above DSL (18.96%) and customers without internet service (7.40%). This may reflect service pricing, reliability, or competition rather than the technology itself, and is investigated further in the modeling stage.

3.6 Churn by Payment Method



BUSINESS INSIGHT — Billing Friction

Electronic-check customers churn at 45.29%, substantially above mailed check (19.11%), automatic bank transfer (16.71%), and automatic credit card (15.24%). This flags electronic-check users as a high-risk segment potentially linked to billing experience or lack of payment automation.

4. Predictive Modeling

Customer identifiers and EDA-only derived fields were removed from the feature set. Data was split 80/20 into training and test sets using stratified sampling to preserve the churn ratio in both subsets. Numerical features were standardized and categorical features one-hot encoded, with all preprocessing fit exclusively on the training data inside a scikit-learn pipeline to prevent information leakage from the test set.

4.1 Model Comparison

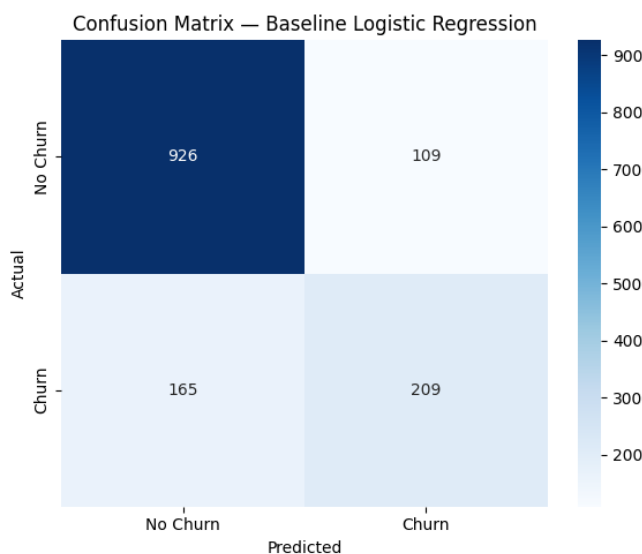
Three classifiers were trained on the identical train/test split and preprocessing pipeline, and evaluated on accuracy, precision, recall, F1-score, and ROC-AUC (the last two given the class imbalance).

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.806	0.657	0.559	0.604	0.842
Random Forest	0.782	0.614	0.481	0.540	0.819
XGBoost	0.805	0.668	0.527	0.589	0.841

BUSINESS INSIGHT — Model Selection

Logistic Regression achieved the strongest overall performance (ROC-AUC 0.842, F1 0.604) and was selected as the working model for its balance of predictive power and interpretability. XGBoost was close behind on ROC-AUC (0.841) and slightly higher on precision (0.668), while Random Forest underperformed both. These models used default/fixed hyperparameters rather than a tuned search, so this ranking should be treated as a first-pass comparison (see Section 8, Limitations).

4.2 Baseline Model Confusion Matrix



The confusion matrix for the Logistic Regression baseline (at the default 0.50 threshold) highlights false negatives — churned customers the model failed to flag — as the key error type to reduce for a retention use case, which motivates the threshold adjustment in Section 5.

5. Business-Oriented Threshold Optimization

By default, classifiers label a customer as “will churn” only when predicted probability exceeds 0.50. For a retention program, missing an actual churner (a false negative) is typically costlier than flagging a loyal customer for unnecessary outreach (a false positive). Several thresholds were therefore evaluated against precision/recall trade-offs on the Logistic Regression model.

Threshold	Precision	Recall	F1-Score
0.30	0.519	0.754	0.615
0.40	0.568	0.668	0.614
0.50 (default)	0.657	0.559	0.604
0.60	0.718	0.401	0.515
0.70	0.742	0.184	0.296

BUSINESS INSIGHT — Selected Threshold: 0.30

Lowering the threshold to 0.30 raises recall from 55.9% to 75.4% and slightly improves F1-score (0.604 → 0.615), at the cost of lower precision (0.519) — more customers will be flagged for retention outreach who would not actually have churned. Given the retention objective, this trade-off was judged acceptable. In production, the exact threshold should be set using the relative cost of a missed churner versus an unnecessary retention offer.

6. Model Explainability

Because Logistic Regression was selected, its coefficients directly indicate which features push predicted churn probability up or down (after standardizing numeric features and one-hot encoding categorical ones). Positive coefficients are associated with higher predicted churn risk; negative coefficients with lower risk.

Feature	Coefficient	Direction
Internet Service: Fiber optic	+0.634	Higher risk
Contract: Month-to-month	+0.583	Higher risk
Total Charges	+0.536	Higher risk
Payment: Electronic check	+0.198	Higher risk
Online Security: No	+0.156	Higher risk
Tech Support: No	+0.132	Higher risk

Tenure	–1.258	Lower risk
Contract: Two year	–0.777	Lower risk
Internet Service: DSL	–0.649	Lower risk
Monthly Charges	–0.592	Lower risk

BUSINESS INSIGHT — Consistency with EDA

The coefficients broadly confirm the EDA findings: fiber-optic service and month-to-month contracts are associated with higher predicted churn, while longer tenure and two-year contracts are associated with lower predicted churn. Note that Total Charges, Monthly Charges, and Tenure are correlated with one another ($\text{Total Charges} \approx \text{tenure} \times \text{Monthly Charges}$), so individual coefficient magnitudes among these three should be read directionally rather than as precise, independent effect sizes. These are model associations, not causal effects.

7. Customer Risk Segmentation & Retention Strategy

Predicted churn probabilities from the Logistic Regression model were converted into three risk tiers to support prioritization of retention effort on the 1,409 test-set customers:

Risk Tier	Probability Range	Customers	Share
Low Risk	0.00 – 0.30	866	61.5%
Medium Risk	0.30 – 0.60	334	23.7%
High Risk	0.60 – 1.00	209	14.8%

These segments are based on model predictions on historical data and should be validated against actual retention outcomes before being used to drive live business decisions.

Recommended Retention Actions

Prioritize early-tenure customers — strengthen onboarding and proactive engagement in the first 6–12 months, where churn risk is highest.

Target month-to-month customers — test incentives for eligible customers to move to longer-term contracts.

Investigate fiber-optic churn — review service quality, pricing, and customer experience before designing targeted interventions for this segment.

Review electronic-check customers — investigate billing/payment friction and promote convenient automatic payment options.

Act on model-identified high-risk customers — prioritize the ~15% of customers above the 0.60 probability mark for direct outreach, with the ~24% above 0.30 considered for broader proactive engagement.

8. Limitations & Future Work

This analysis is a first-pass predictive model rather than a production-ready system. The following points should be addressed before deployment:

Threshold selection on the test set. The 0.30 decision threshold was chosen by evaluating precision/recall directly on the test set, which was also used for final model evaluation. This risks an optimistic estimate of real-world performance. A separate validation split (or nested cross-validation) should be used to select the threshold, reserving the test set purely for a final, untouched check.

No cross-validation. All model comparisons rely on a single 80/20 split. Metrics such as ROC-AUC (0.842 vs. 0.841 vs. 0.819) are close enough between models that split-to-split variance could affect the ranking; k-fold cross-validation would give more reliable comparisons.

Limited hyperparameter tuning. Random Forest and XGBoost were run with fixed, untuned hyperparameters. A tuned search (grid or randomized search with cross-validation) may change the model ranking reported in Section 4.

Class imbalance handling. The current approach relies on threshold adjustment rather than resampling (e.g. SMOTE) or class-weighting; both are worth testing as alternatives or complements.

Validation against real outcomes. Risk segments and recommended actions are based on model predictions on historical data and have not yet been validated against actual retention campaign results.

Tech stack: Python, pandas, scikit-learn, XGBoost, matplotlib, seaborn.

Source: Telco Customer Churn dataset (IBM / Kaggle). This report was generated from the accompanying Jupyter notebook — all metrics and charts reflect the notebook's executed output.